Magnetostatics

Problem 5.1

Problem 5.2

Problem 5.3

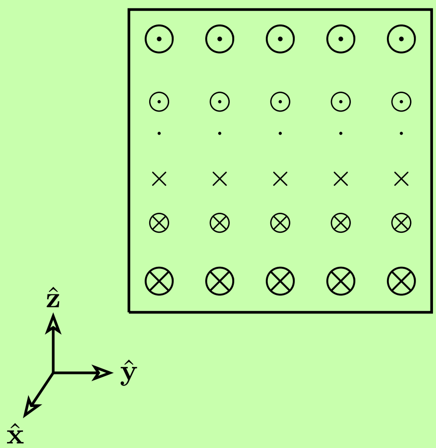
Problem 5.4

Suppose that the magnetic field in some region has the form

$$B = kz\hat{\mathbf{x}}$$

(where k is a constant). Find the force on a square loop (side a), lying in the yz plane and centered at the origin, if it carries a current I , flowing counterclockwise, when you look down the x axis.

Solution:



The force on a wire carrying a current $\mathbf{I}$ is

$$\mathbf{F} = \int I(\mathrm{dl} \times \mathbf{B})$$

With current I, the forces on each of the four lines are, starting with the top line:

$$z = \frac{a}{2}, \quad y : \frac{a}{2} \to -\frac{a}{2}$$
$$B = k\frac{a}{2}\hat{\mathbf{x}}, \quad \mathrm{dl} = \hat{\mathbf{y}}\,\mathrm{d}y$$
$$\mathbf{F} = Ik\frac{a}{2}\int_{\frac{a}{2}}^{-\frac{a}{2}} (\hat{\mathbf{y}} \times \hat{\mathbf{x}})\,\mathrm{d}y = -Ik\frac{a}{2}\hat{\mathbf{z}}\int_{\frac{a}{2}}^{-\frac{a}{2}} \mathrm{d}y = Ik\frac{a}{2}\hat{\mathbf{z}}\int_{-\frac{a}{2}}^{\frac{a}{2}} \mathrm{d}y$$
$$\downarrow$$
$$\mathbf{F} = \frac{Ika^2}{2}\hat{\mathbf{z}}$$

Bottom line:

$$B = -k\frac{a}{2}\hat{\mathbf{x}}, \quad \mathrm{dl} = \hat{\mathbf{y}}\,\mathrm{d}y, y : -\frac{a}{2} \to \frac{a}{2}$$
$$\mathbf{F} = -Ik\frac{a}{2}\int_{-\frac{a}{2}}^{\frac{a}{2}} (\hat{\mathbf{y}} \times \hat{\mathbf{x}})\,\mathrm{d}y = Ik\frac{a}{2}\hat{\mathbf{z}}\int_{-\frac{a}{2}}^{\frac{a}{2}} \mathrm{d}y$$
$$\downarrow$$
$$\mathbf{F} = \frac{Ika^2}{2}\hat{\mathbf{z}}$$

The forces on the left and right wires are zero due to symmetry (there is equal parts of these wires in a $+\hat{\mathbf{x}}$ magnetic field as $-\hat{\mathbf{x}}$ magnetic field).

Total is top + Bottom

$$\mathbf{F}_{\text{tot}} = \mathbf{F}_{\text{top}} + \mathbf{F}_{\text{bot}} = \frac{Ika^2}{2}\hat{\mathbf{z}} + \frac{Ika^2}{2}\hat{\mathbf{z}} = \boxed{Ika^2\hat{\mathbf{z}}}$$

□

Problem 5.5

A current I flows down a wire of radius a.

(a) If it is uniformly distributed over the surface, what is the surface current density K ?

(b) If it is distributed in such a way that the volume current density is inversely proportional to the distance from the axis, what is $J(s)$?

Solution:

(a) A wire of radius a has circumference $2\pi a$ so the current passing through each infitesimal part of the surface of the wire (circumference) is simply

$$K = \frac{I}{2\pi a}$$

(b) We want the volume current density to be

$$J = \frac{k}{s}$$

where k is the proportionality constant. If we were to integrate $J(s)$ over a cross section of the wire, we should recover I

$$I = \int J(s) = \int_0^a \int_0^{2\pi} \frac{k}{s} s \, \mathrm{d}s \, \mathrm{d}\phi = \int_0^a \int_0^{2\pi} k \, \mathrm{d}s \, \mathrm{d}\phi \to I = ka2\pi$$

$$\downarrow$$

$$k = \frac{I}{2\pi a}$$

so

$$J(s) = \frac{I}{2\pi as}$$

□

Problem 5.6

Problem 5.7

Problem 5.8

(a) Find the magnetic field at the center of a square loop, which carries a steady current I . Let R be the distance from center to side.

(b) Find the field at the center of a regular n-sided polygon, carrying a steady current I. Again, let R be the distance from the center to any side.

(c) Check that your formula reduces to the field at the center of a circular loop, in the limit $n \to \infty$.

Solution:

(a) We can find the field produced by one of the sides, and due to symmetry the result is four times of that. Calculating the right wire gives

$$
\begin{gathered}
x = R, \quad y : -R \to R, \quad \mathrm{d}\mathbf{l}' = \hat{\mathbf{y}}\,\mathrm{d}y \\
\boldsymbol{\mathscr{r}} = -R\hat{\mathbf{x}} - y\hat{\mathbf{y}}, \quad \mathscr{r} = \sqrt{R^2 + y^2} \\
\hat{\boldsymbol{\mathscr{r}}} = \frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}}, \quad \mathrm{d}\mathbf{l}' \times \hat{\boldsymbol{\mathscr{r}}} = \frac{R}{\sqrt{R^2 + y^2}^3}\,\mathrm{d}y\hat{\mathbf{z}}
\end{gathered}
$$

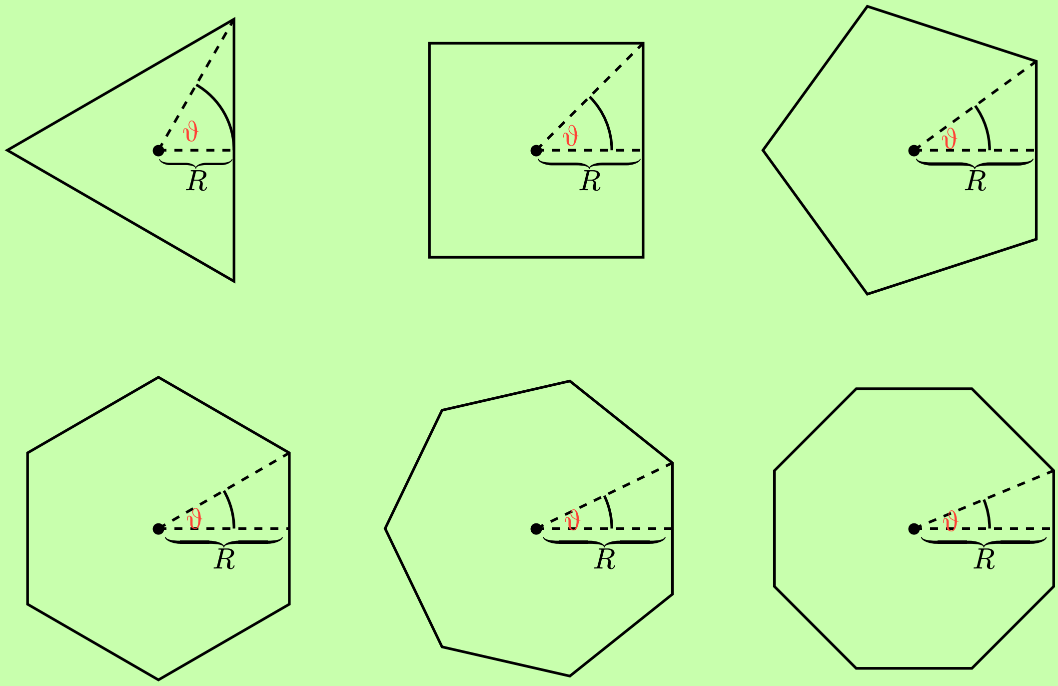
Using these expressions, Biot-Savart law becomes

$$
\begin{aligned}
\mathbf{B} = \frac{\mu_0 I}{4\pi}\hat{\mathbf{z}} \int_{-R}^{R} \frac{R}{\sqrt{R^2 + y^2}^3}\,\mathrm{d}y = \frac{\mu_0 I}{4\pi}\hat{\mathbf{z}} \left[\frac{Ry}{R^2\sqrt{R^2 + y^2}} \right]_{-R}^{R} = \\
= \frac{2\mu_0 I}{4\pi R\sqrt{2}}\hat{\mathbf{z}} = \frac{\sqrt{2}\mu_0 I}{4\pi R}\hat{\mathbf{z}}
\end{aligned}
$$

Multiplying this result with 4 (since there are 4 straight wire segments) gives

$$
\mathbf{B}(\mathbf{0}) = \frac{\sqrt{2}\mu_0 I}{\pi R}\hat{\mathbf{z}}
$$

(b) For a n-sided polygon, the same principle holds, where if we can find the field from one of the straight segments, the total field is given by multiplying this result with n.



The total length of the straight right edges for all polygons is

$$L = 2R\tan(\theta), \quad \theta = \frac{\pi}{n}$$

so Biot-Savart becomes

$$\frac{R\mu_0 I}{4\pi}\hat{\mathbf{z}}\int_{-R\tan\left(\frac{\pi}{n}\right)}^{R\tan\left(\frac{\pi}{n}\right)} \frac{1}{\sqrt{R^2+y^2}^3}\,\mathrm{d}y = \frac{R\mu_0 I}{4\pi}\left[\frac{y}{R^2\sqrt{R^2+y^2}}\right]_{-R\tan\left(\frac{\pi}{n}\right)}^{R\tan\left(\frac{\pi}{n}\right)} =$$

$$= \frac{\mu_0 I\tan\left(\frac{\pi}{n}\right)}{2\pi R\sqrt{1+\tan^2\left(\frac{\pi}{n}\right)}}\hat{\mathbf{z}}$$

Now, $\tan(\theta) = \sin(\theta)/\cos(\theta)$ and $\sqrt{1+\tan^2(\theta)} = \sec(\theta) = 1/\cos(\theta)$, so the field from the right line segment is

$$\mathbf{B} = \frac{\mu_0 I\sin\left(\frac{\pi}{n}\right)}{2\pi R}\hat{\mathbf{z}}$$

Multiplying by the amount of sides for the polygon, n, gives the total magnetic field

$$\mathbf{B} = \frac{\mu_0 I}{2R}\frac{\sin\left(\frac{\pi}{n}\right)n}{\pi}\hat{\mathbf{z}} = \frac{\mu_0 I}{2R}\frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}}\hat{\mathbf{z}}$$

(c) As $n \to \infty$, the term $\frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \to 1$ (since $\frac{\pi}{n} \to 0$ as $n \to \infty$, and we have the standard limit $\frac{\sin(x)}{x} \to 1$ as $x \to 0$). This gives the magnetic field

$$\mathbf{B} = \frac{\mu_0 I}{2R}\hat{\mathbf{z}}$$

which is exactly the field given by a circular loop of radius R.

$\square$

Problem 5.9

Problem 5.10

Problem 5.11

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